Benchmarking and Adapting Open-Source Vision-Language Models for TextVQA: A Funnel Evaluation and LoRA Fine-Tuning Study

Chenxi Ge
Harvard University
School of Engineering and Applied Sciences
steven_ge@fas.harvard.edu

Harry Hu
Harvard University
School of Engineering and Applied Sciences
harryhu@fas.harvard.edu

Abstract

TextVQA requires a model to read scene text, ground it in visual context, and answer short natural-language questions under exact-match evaluation. We present a staged empirical study that benchmarks open-source vision-language models (VLMs) under a funnel protocol designed to preserve validation integrity while still supporting broad exploration. The protocol combines stratified internal-dev screening, finalist reruns on the official validation split, winner-conditioned LoRA fine-tuning, and post-training adapter evaluation under a reproducible local-plus-remote execution boundary. Across four open-source VLM families and OCR-aware prompting variants, Qwen2.5-VL-3B emerges as the clear zero-shot winner, reaching 81.05% internal-dev accuracy and 78.72% on official validation with a short-answer prompt. A subsequent 12-run LoRA study raises the final official validation score to 84.76%, a paired gain of 6.04 accuracy points over the strongest zero-shot finalist, with parallel gains in consensus accuracy, F1, and ROUGE-L. The trained-model analysis further shows that adaptation gains are concentrated in OCR-grounded cases rather than OCR-free questions, while follow-up runs reveal only marginal sensitivity to OCR toggling after adaptation and a modest gain from scaling the training pool. Beyond the headline result, the paper contributes a defensible evaluation funnel, a compute-aware multi-stage study design, and a reproducible artifact stack that supports quantitative, qualitative, and robustness-oriented analysis rather than a single isolated benchmark number.

1 Introduction

Reading text in natural images remains a central failure mode for otherwise capable vision-language systems. The TextVQA benchmark was introduced precisely to stress this gap: a successful model must not only parse scene text, but also integrate it with visual context and question semantics to produce a short, exact answer [Singh et al., 2019]. In practice, this setting is harder than generic image question answering [Antol et al., 2015] because many correct responses depend on OCR fidelity, prompt format, answer normalization, and the model's ability to reason over text that is sparse, noisy, or visually embedded in cluttered scenes.

At the same time, the open-source VLM ecosystem has matured enough that a serious empirical study is now feasible without closed commercial models. Architectures such as BLIP-2 [Li et al., 2023], LLaVA [Liu et al., 2023], InternVL2.5 [Chen et al., 2024], and Qwen2.5-VL [Qwen Team, 2025] offer strong multimodal priors at parameter scales that can still be evaluated and adapted in an academic setting. Yet the existence of capable models does not, by itself, solve the experimental-design problem. A naïve study can easily become either too shallow, by comparing only a small number of zero-shot settings, or too diffuse, by mixing screening, validation, prompt exploration,

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and fine-tuning without a clear promotion policy. In TextVQA, where OCR choices matter almost as much as backbone choice, such looseness makes the final conclusions difficult to defend.

This paper therefore adopts a deliberately staged protocol rather than a flat benchmark. We first run broad zero-shot screening on a stratified internal development split, then reserve the official validation set for a narrower set of finalists, and finally conduct a winner-conditioned LoRA study on the selected backbone. The key principle is that each stage should answer a different scientific question: screening compares backbones and OCR-sensitive prompting cheaply, finalist reruns verify which combinations are still competitive on the official split, and the training stage isolates the effect of parameter-efficient adaptation while keeping follow-up ablations attached to a selected LoRA family rather than to an arbitrary default. The final claims are made on official validation evidence, while internal-dev results are used for screening, diagnostics, and budget allocation. The result is an experiment that is broader than a single-tuned-model class project, but still structured enough to remain methodologically coherent.

Our study is also designed around realistic compute constraints rather than idealized cluster access. All canonical evaluation runs are executed locally in a resumable queue on a single Apple-Silicon workstation, while only the final LoRA matrix is moved to rented remote GPUs under a separate run namespace. This separation is not merely operational. It preserves a clean experimental boundary between the local zero-shot/finalist evidence and the remote training evidence, reducing the risk that nominally identical runs mix hardware regimes, caches, or checkpoints in ways that complicate interpretation.

The paper makes the following contributions. First, it introduces a reproducible funnel protocol for benchmarking open-source VLMs on TextVQA while preserving the official validation split for finalist and final tuned-model evidence. Second, it treats OCR as a first-class experimental variable across zero-shot prompting, training-time ablation, and post-hoc robustness analysis. Third, it develops a winner-conditioned LoRA study in which training-stage follow-up allocation, internal-dev adapter promotion, and official-validation claims are explicitly separated. This separation keeps a local diagnostic objective distinct from the final paper-facing evaluation objective. Finally, the project delivers a complete implementation pipeline that couples layered configuration, resumable execution, remote multi-GPU orchestration, progress reporting, and generated audit artifacts, making the empirical claims auditable rather than anecdotal.

Section 2 situates the study with respect to TextVQA, open-source VLMs, and parameter-efficient adaptation. Section 3 then formalizes the staged experimental protocol and systems boundary that govern the full study.

2 Related Work

2.1 TextVQA and text-grounded visual question answering

TextVQA was introduced to measure a model's ability to answer questions whose solution depends on reading text embedded in real-world images [Singh et al., 2019]. It extends an earlier line of work on generic visual question answering [Antol et al., 2015] and scene-text-centered variants such as ST-VQA [Biten et al., 2019]. Unlike generic VQA, the benchmark frequently requires exact recognition of sparse text, reasoning over OCR tokens, and answer generation under strict string-matching evaluation. This makes the problem particularly sensitive to small prompt decisions, OCR formatting, and answer normalization. For an empirical paper, the implication is straightforward: a convincing TextVQA study must treat text handling as part of the research question rather than as an implementation footnote.

Earlier TextVQA systems made this dependency explicit by combining visual features with OCR tokens and copy-style answer mechanisms. The LoRRA baseline introduced with TextVQA already showed that generic VQA models leave substantial performance on the table when scene text is needed [Singh et al., 2019], and M4C later used pointer-augmented multimodal transformers to reason over OCR tokens directly [Hu et al., 2020]. Modern instruction-tuned VLMs no longer require the same task-specific architecture to produce plausible answers, but the underlying evaluation issue remains: the experimenter must still decide how text is exposed to the model, how exact answers are normalized, and how much validation evidence is spent during prompt exploration.

That requirement shapes the present study in two ways. First, OCR-aware prompting is treated as a structured experimental variable throughout zero-shot evaluation rather than as a single fixed preprocessing choice. Second, OCR remains part of the fine-tuning analysis through explicit on/off follow-up runs and post-hoc bucketed validation analysis. This design choice is motivated by the benchmark itself: if scene text is central to the task, then conclusions about backbone quality or adaptation quality are incomplete unless OCR usage is interrogated directly.

2.2 Open-source vision-language models

Open-source VLMs have advanced rapidly through increasingly effective combinations of pretrained vision encoders, pretrained language models, and instruction-tuning pipelines. BLIP-2 demonstrated that strong multimodal transfer can be achieved by bootstrapping from frozen unimodal components rather than retraining the full stack from scratch [Li et al., 2023]. LLaVA showed that visual instruction tuning can substantially improve general-purpose multimodal behavior with a comparatively lightweight alignment stage [Liu et al., 2023]. More recent model families, including Qwen2.5-VL [Qwen Team, 2025] and InternVL2.5 [Chen et al., 2024], have pushed open-source multimodal performance far beyond early proof-of-concept systems, making it realistic to benchmark several families within one study.

However, strong checkpoints alone do not guarantee a rigorous comparison. Public model reports typically emphasize aggregate benchmark leaderboards, while a project like ours must instead decide how to compare models under a fixed, transparent protocol. In particular, an academic benchmark on TextVQA must choose how much prompt breadth to allow, which split should be used for screening, and when it is appropriate to spend the official validation budget. Our funnel design can be understood as a methodological answer to that problem: broad internal-development screening is used to expose prompt and backbone sensitivity, while the official validation split is deliberately held back for a much smaller set of finalists and final tuned-model checks.

2.3 Parameter-efficient adaptation

The fine-tuning stage in this project is built around Low-Rank Adaptation (LoRA) [Hu et al., 2022], which remains the most practical parameter-efficient adaptation method for a study that aims to compare several adaptation settings rather than optimize a single maximal model. LoRA is especially appropriate here for three reasons. First, it makes a multi-run ablation feasible within a limited compute budget. Second, it allows adaptation choices such as target-module strategy and rank to be varied systematically. Third, it keeps the training question close to the paper's actual scope: we are not trying to establish a new state of the art on TextVQA, but rather to understand how far a strong open-source backbone can be pushed under a controlled and reproducible adaptation budget.

Most importantly, we treat LoRA not as a one-off fine-tuning step but as a structured study. The core matrix compares attention-only and all-linear target strategies, rank 16 versus 32, and seed stability. The follow-up runs then ask higher-level questions that the core matrix alone cannot answer: whether OCR conditioning materially changes adapted performance, and whether additional data continues to improve the selected family. This winner-conditioned design is more informative than a single tuned checkpoint and more defensible than ablations attached to a family that never actually won the initial training comparison.

3 Study Design and Experimental Protocol

3.1 Task and data protocol

The study follows the official TextVQA split structure of approximately 34.6k training examples, 5k validation examples, and 5.73k test examples [Singh et al., 2019]. We reserve the official validation split for finalist reruns, appendix experiments, and final tuned-model checks. Because public labels for the official test split are not available in the local dataset release used by this project, all paper-facing final claims are made on the official validation split. To avoid using that validation split for broad exploratory screening, we materialize a separate $2{,}000$-question internal development set from the training split. This internal-dev set is stratified rather than sampled naively, with explicit control over question prefix, answer-length bucket, and OCR-token-count bucket. The remaining training questions form the training pool for the LoRA study.

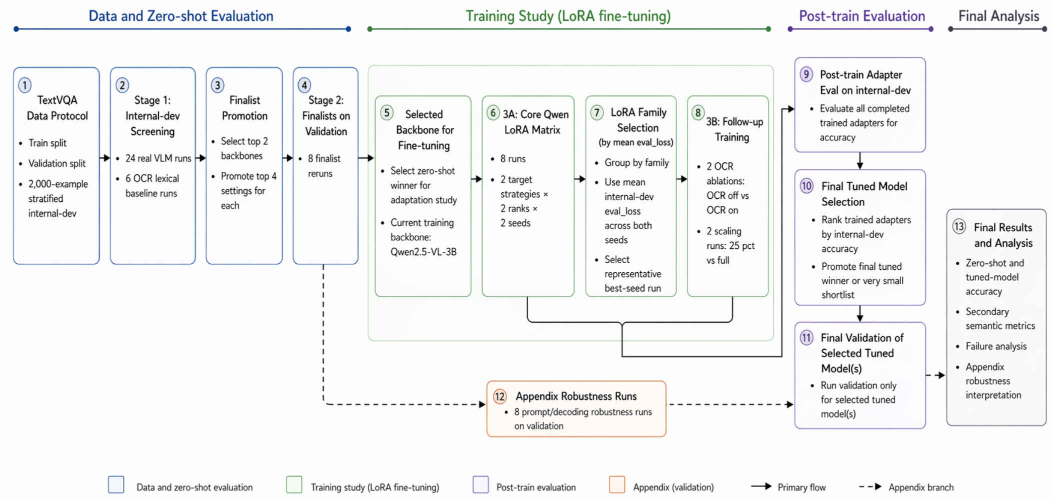

Figure 1: End-to-end experiment pipeline for the current TextVQA study. The protocol follows a staged funnel: stratified internal-dev screening, finalist reruns on the official validation split, backbone selection for LoRA fine-tuning, an 8-run core Qwen LoRA matrix, and 4 winner-conditioned follow-up training runs. Training follow-ups are chosen by mean internal-dev `eval_loss` across core seeds, whereas completed trained adapters are later ranked by internal-dev accuracy before promoting only the final tuned model, or a very small shortlist, to validation. Appendix robustness runs remain a secondary validation branch that supports the final analysis without altering the main winner-selection funnel.

Pool	QA	Images	Image relation	Role
Internal-dev	2000	1939	571 image-disjoint QA; 1429 image-overlap QA	Screening/diagnostics
Train remainder	32602	21443	Question-disjoint from internal-dev	LoRA training pool
Official validation	5000	3166	Official held-out split	Final paper-facing evaluation

Table 1: Data-use audit for the current study. Internal-dev is useful for inexpensive screening and adapter diagnostics, but official validation is the final paper-facing evaluation split.

The internal-dev split is question-disjoint from the LoRA training pool, but it is not fully image-disjoint: TextVQA contains multiple questions for some images, and the current split removes selected questions rather than every question attached to the same image. Table 1 makes this boundary explicit. We therefore treat internal-dev results as screening and diagnostic evidence rather than as independent final test evidence. The final claims in the paper are anchored on the official validation split, and Section 5 reports an overlap sensitivity check to show how much the internal-dev adapter ranking changes when only image-disjoint internal-dev questions are retained.

This split policy serves two methodological goals. First, it allows broad model and prompt exploration without spending the official validation split on every exploratory setting. Second, it makes the final validation comparisons more meaningful, because they are reserved for a shortlist that has already survived a cheaper but structured screening stage. The protocol therefore trades a small amount of up-front split engineering for a cleaner evidence hierarchy.

3.2 Model pool and prompt settings

The canonical zero-shot benchmark pool contains four open-source VLM families and one non-neural OCR lexical baseline. The four reportable VLM families are Qwen2.5-VL-3B [Qwen Team, 2025], InternVL2.5-4B [Chen et al., 2024], LLaVA-Phi-3-mini [Liu et al., 2023], and BLIP-2 OPT-2.7B [Li et al., 2023]. The OCR lexical baseline is included not as a competitor to modern VLMs, but as a calibration point that helps distinguish genuine multimodal reasoning from question-answer overlap driven by visible text alone.

Each real VLM is screened under six prompt settings: `plain`, `short_answer`, `ocr_copy_first`, `ocr_injected`, `ocr_injected_normalized`, and `ocr_fused`. This yields 24 real VLM screening runs (4 backbones $\times$ 6 settings) plus 6 OCR-baseline runs. The design deliberately treats OCR prompting as a first-class variable, because TextVQA performance is often determined as much by how text is exposed to the model as by which backbone is chosen.

3.3 Funnel evaluation protocol

The first evaluation stage performs broad screening on the internal-dev split. Its purpose is not to produce final leaderboard claims, but to estimate which backbone–prompt combinations are competitive enough to deserve the more expensive validation budget. From this stage, we promote the top two backbones and the top four settings for each, producing eight finalist reruns on the official validation split. A separate appendix branch contains eight additional prompt and stress-test evaluations that are intended to strengthen the analysis section rather than alter the main winner-selection logic.

This funnel design is central to the paper's methodology. It prevents the study from becoming either too small to be informative or too loose to be interpretable. In particular, it allows prompt sensitivity to be measured early, but avoids the common mistake of reporting many exploratory validation runs as if they carried the same evidentiary weight as a small finalist set. The backbone selected for fine-tuning is therefore not the winner of a single flat leaderboard, but the survivor of a staged promotion policy.

3.4 Winner-conditioned LoRA training study

The adaptation path in this project is LoRA fine-tuning on Qwen2.5-VL-3B. The training study has twelve runs in total. The core matrix contains eight runs spanning two target-module strategies, two ranks, and two random seeds. Concretely, it compares attention-only versus all-linear target sets and rank 16 versus 32, with seed replication to estimate whether observed differences are stable or fragile.

The remaining four runs are follow-ups conditioned on the completed core matrix rather than fixed in advance to an arbitrary default family. The first pair is an OCR-conditioning ablation that toggles OCR on and off while holding the selected LoRA family fixed. The second pair is a data-scaling comparison between a smaller fixed-budget subset and the full training-remainder pool. We deliberately separate three decisions that are easy to conflate. Mean internal-dev `eval_loss` across the two seeds is used only as a budget-allocation signal for choosing which LoRA family receives follow-up OCR and scaling runs. Internal-dev answer accuracy is then used to choose the single trained adapter promoted to official validation. The paper-facing final claim is made only after that adapter is evaluated on the official validation split.

This separation between training-stage loss, internal-dev accuracy, and official-validation accuracy is intentional. The first is a control signal for allocating follow-up training budget, the second is a local promotion rule, and the third is the final evaluation criterion. Keeping these roles distinct makes the adaptation study easier to interpret and prevents diagnostic follow-ups from being mistaken for the final model-selection evidence.

3.5 Metrics, systems boundary, and reproducibility

The headline metric is exact any-match accuracy after TextVQA-style answer normalization. A prediction receives `Acc.`=1 if it exactly matches at least one of the ten human reference answers after normalization and 0 otherwise. We also report `Cons.`, the consensus score $\min(m/3, 1)$, where m is the number of normalized human answers that exactly match the prediction. The former is the primary paper-facing score because it directly tracks whether the model produced an acceptable short answer; the latter is included because it preserves the conventional VQA intuition that answers supported by more annotators should receive more credit.

In addition, we report secondary semantic and overlap metrics that capture partial-answer quality beyond exact string matching, including BLEU [Papineni et al., 2002], METEOR [Banerjee and Lavie,

Component	Setting
Backbone	Qwen/Qwen2.5-VL-3B-Instruct
Training dtype / device	bfloat16 on remote NVIDIA H100 GPUs
Core training budget	$8{,}192$ training questions, 1.0 epoch, effective batch size 8
Core learning rates	2×10^{-4} for rank 16; 1.5×10^{-4} for rank 32
Optimizer schedule	warmup ratio 0.03, weight decay 0.0
LoRA modules	`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`
LoRA regularization	$\alpha = 2r$, dropout 0.05, bias disabled
Training prompt	`ocr_injected_normalized`, dataset OCR, max 32 OCR tokens
Zero-shot finalist prompt	`short_answer`, no explicit OCR injection

Table 2: Key implementation settings for the Qwen LoRA study and the strongest zero-shot finalist. The full run configuration is materialized in the saved `settings.json` files and layered TOML configs.

2005], ROUGE-L [Lin, 2004], token-level F1, precision/recall, and an LLM-as-a-Judge similarity score across all paper-reported experiment tables. The judge score is computed post hoc from saved prediction files using GPT-4.1 mini via the OpenAI API [OpenAI, 2025] and follows the broader practice of LLM-based semantic adjudication in evaluation pipelines [Zheng et al., 2023]. Each prediction is compared against the original question and the full set of human reference answers and receives a discrete semantic score in $\{0, 0.5, 1\}$, where 1 denotes semantic equivalence to at least one acceptable answer, 0.5 denotes a substantial but incomplete near match, and 0 denotes a semantic mismatch. The judge score is used only as a secondary diagnostic; it is never used to select the final model or replace the benchmark-native exact metrics.

The implementation boundary is equally deliberate. All canonical zero-shot, finalist, and appendix evaluations are executed locally on a single Apple-Silicon workstation with resumable queueing and append-only logs. The LoRA training matrix and trained-adapter evaluations are run on remote H100 GPUs under a distinct run namespace rather than mixed with earlier local pilot training attempts. This preserves a clean hardware and checkpoint boundary between the local zero-shot/finalist evidence and the remote adaptation evidence. Reproducibility is further supported through layered TOML configuration, materialized manifests, resumable training checkpoints, automated progress reporting, and remote multi-GPU orchestration with explicit phase boundaries. The result is an experiment that is not only ambitious in scope, but also inspectable in its operational details.

4 Results

4.1 Zero-shot screening and finalist reruns

The internal-dev screening matrix immediately separates backbone quality from prompt sensitivity. Figure 2 shows that Qwen2.5-VL-3B dominates the full screening stage, with its best configuration (`short_answer`) reaching 81.05% accuracy and its weakest OCR-heavy variant still remaining competitive. The ranking is not a trivial prompt artifact: even after allowing six prompt/OCR settings per backbone, Qwen remains clearly ahead of LLaVA-Phi-3-mini, BLIP-2 OPT-2.7B, InternVL2.5-4B, and the OCR lexical baseline. The same figure also shows that prompt sensitivity is model dependent. Qwen prefers concise answer formatting, whereas LLaVA benefits more from OCR-normalized prompting; by contrast, BLIP-2 remains weak across all prompt variants, and InternVL2.5-4B never becomes competitive under the study protocol.

The finalist stage confirms that the screening winner transfers to the official validation split. Table 3 shows that the four promoted Qwen settings occupy the top four validation positions, led again by `short_answer` at 78.72% accuracy and 89.09% judge similarity. The strongest non-Qwen finalist, LLaVA-Phi-3-mini with OCR-normalized prompting, reaches only 50.36% exact-match accuracy, leaving a 28.36-point gap to the best Qwen finalist. This margin is large relative to the within-family prompt differences, so under the current evaluation boundary the main zero-shot ranking is better explained by backbone quality than by any single OCR-aware prompt variant.

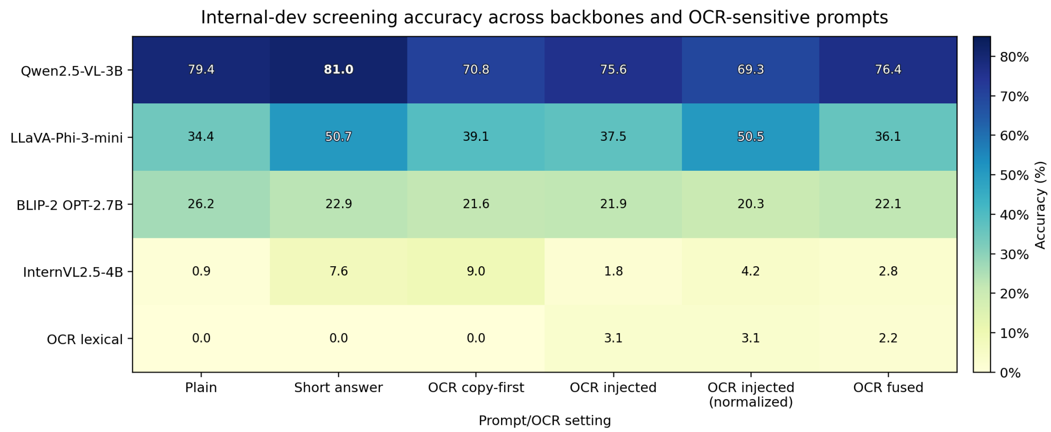

Figure 2: Internal-dev screening accuracy across the full backbone–prompt matrix. Qwen2.5-VL-3B is the only family that remains strong across all OCR-sensitive prompt variants, and its short-answer prompt is the best zero-shot configuration overall. The heatmap also makes the model-dependent nature of OCR prompting visible: OCR normalization helps LLaVA markedly more than it helps Qwen, while the lexical OCR baseline remains near zero throughout.

Stage	Model / setting	Acc.	Cons.	F1	BLEU	METEOR	ROUGE-L	Judge
Best screening	Qwen2.5-VL-3B (short answer)	81.05	77.20	70.24	13.48	51.26	72.38	90.95
Best screening	LLaVA-Phi-3-mini (short answer)	50.70	46.93	45.74	7.29	36.24	48.03	71.58
Best screening	BLIP-2 OPT-2.7B (plain)	26.25	23.50	23.62	2.58	25.13	24.83	59.10
Best screening	InternVL2.5-4B (ocr copy first)	9.05	8.78	11.84	0.14	8.26	19.77	55.53
Best screening	OCR lexical (ocr injected)	3.15	2.83	10.45	0.59	11.00	11.43	36.98
Finalist	Qwen2.5-VL-3B (short answer)	78.72	74.45	70.42	**14.88**	48.08	73.45	89.09
Finalist	Qwen2.5-VL-3B (plain)	77.18	72.91	69.47	14.64	47.81	72.78	87.69
Finalist	Qwen2.5-VL-3B (ocr fused)	75.56	70.93	67.37	14.14	46.88	70.84	85.87
Finalist	Qwen2.5-VL-3B (ocr injected)	74.42	69.99	66.69	13.92	45.93	70.26	85.46
Finalist	LLaVA-Phi-3-mini (ocr injected normalized)	50.36	46.98	48.72	7.26	35.49	52.34	73.77
Finalist	LLaVA-Phi-3-mini (short answer)	46.70	43.42	45.08	7.68	33.06	47.99	72.19
Finalist	LLaVA-Phi-3-mini (ocr injected)	40.68	37.65	41.66	5.13	32.26	46.80	69.40
Finalist	LLaVA-Phi-3-mini (ocr copy first)	36.86	34.42	41.81	7.07	32.97	44.67	68.11
Tuned winner	Qwen2.5-VL-3B + LoRA (all-linear r16 seed13)	**84.76**	**80.39**	**74.92**	14.26	**50.66**	**77.56**	**91.24**

Table 3: Main benchmark results. The table reports the best internal-dev screening configuration for each backbone, all promoted validation finalists, and the final tuned model. Exact-match accuracy is the primary score; consensus accuracy, F1, BLEU, METEOR, ROUGE-L, and judge similarity provide secondary semantic context. Bold entries mark the best value among validation-facing rows only; the internal-dev screening rows are not bolded against validation rows.

4.2 Winner-conditioned LoRA adaptation

The adaptation stage improves the official validation result, not only the internal diagnostic split. Figure 3 reports the headline comparison: the final tuned model, `all-linear-r16-seed13`, reaches 84.76% validation accuracy, compared with 78.72% for the strongest zero-shot finalist, for an absolute improvement of 6.04 points. The same comparison also improves consensus accuracy by 5.95 points, F1 by 4.50 points, and ROUGE-L by 4.11 points. Because both systems are evaluated on the same 5,000 validation questions, we also run a paired bootstrap over examples. The paired 95% confidence interval for the accuracy gain is $[5.00, 7.04]$ points, and an exact McNemar test on the discordant decisions gives $p = 1.97 \times 10^{-32}$.

The semantic judge agrees with the direction of the exact-match result while tightening its interpretation. In Table 3, the strongest zero-shot finalist reaches 89.09% judge similarity, whereas the tuned winner reaches 91.24%, a gain of 2.15 points. This gap is smaller than the exact-match improvement, which is exactly the pattern we would expect if part of the tuned model's advantage comes from stricter normalization and answer-format control rather than from a wholesale change in semantic competence. In other words, the judge score does not erase the tuning gain; it refines it into a more conservative but still positive semantic improvement.

System	Acc. [95% CI]	Cons.	Judge
Zero-shot Qwen finalist	78.72 [77.60, 79.88]	74.45	89.09
Tuned Qwen LoRA	**84.76 [83.86, 85.76]**	**80.39**	**91.24**
Paired gain	6.04 [5.00, 7.04]	–	–

Table 4: Official validation headline comparison. Bold entries mark the stronger system-level metric. The paired gain is computed over the same 5,000 validation questions and is therefore a stronger test than comparing independent intervals alone.

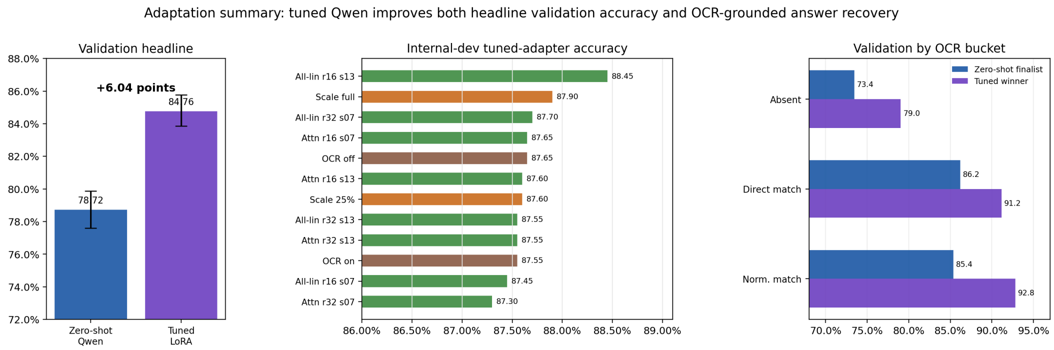

Figure 3: Adaptation summary. Left: the final tuned Qwen model improves validation accuracy by 6.04 points over the strongest zero-shot finalist. Middle: internal-dev post-train accuracy across the full 12-run LoRA study, including the core matrix and winner-conditioned follow-ups. Right: validation gains are largest for OCR-grounded answers, especially those that require normalized OCR matching.

The internal-dev adapter sweep in Figure 3 and Table 5 also reveals why the selection policy separates training loss from answer accuracy. Mean internal-dev `eval_loss` selected the `all-linear-r32` family for the follow-up OCR and scaling diagnostics, but the highest post-train internal-dev answer accuracy among the completed core adapters came from `all-linear-r16-seed13` at 88.45%. This is not a conflict between two winners; it is a distinction between two roles. Loss is used to allocate limited follow-up training budget, while answer accuracy selects the adapter promoted to official validation.

Seed stability in the core matrix is also stronger than the raw leaderboard might suggest. Averaging the two seeds within each family, `all-linear-r16` reaches 87.95% internal-dev accuracy, compared with 87.63% for `all-linear-r32`, 87.63% for `attn-r16`, and 87.43% for `attn-r32`. More importantly, the seed ranges are small: 1.00 point for `all-linear-r16`, 0.15 for `all-linear-r32`, 0.05 for `attn-r16`, and 0.25 for `attn-r32`. The study therefore does not hinge on a single anomalous seed.

The follow-up runs sharpen the interpretation of the core matrix rather than merely extending it. The OCR ablation pair differs by only 0.10 internal-dev accuracy points (87.65% without OCR conditioning versus 87.55% with OCR conditioning), suggesting that once the model is adapted, explicit OCR toggling matters less than it did in zero-shot prompting. The data-scaling pair is more informative: moving from the 25% setting to the full training-remainder run raises internal-dev accuracy from 87.60% to 87.90%. This gain is modest and diagnostic; it suggests that additional data remains useful for the selected follow-up family, but it is not used to override the accuracy-based validation promotion rule.

4.3 Appendix robustness runs

The appendix branch strengthens the analysis without altering the main winner-selection funnel. Table 6 shows that the strongest appendix configuration, `question_routed`, is competitive under both exact-match accuracy and judge similarity, but it remains below the finalist-stage `short_answer` configuration, which is exactly the role that the appendix branch should play: it offers stress tests and

Group	Configuration	Eval loss	Acc.	Cons.	F1	METEOR	ROUGE-L	Judge
Core matrix	All-linear r16 (seed 13)	0.5759	**88.45**	**85.03**	76.72	55.05	**78.34**	92.22
Data scaling	Data scaling: full	**0.5576**	87.90	84.80	76.71	54.25	78.22	**93.83**
Core matrix	All-linear r32 (seed 07)	0.5654	87.70	83.92	75.96	54.77	77.44	93.17
Core matrix	Attention-only r16 (seed 07)	0.5787	87.65	84.05	76.48	54.92	77.93	92.05
OCR ablation	OCR ablation: off	0.5715	87.65	84.13	76.10	**55.19**	77.64	93.35
Core matrix	Attention-only r16 (seed 13)	0.5769	87.60	84.38	76.49	54.61	78.04	92.45
Data scaling	Data scaling: 25%	0.5724	87.60	84.02	75.98	54.71	77.57	92.42
Core matrix	All-linear r32 (seed 13)	0.5833	87.55	84.42	76.22	54.06	77.73	92.47
Core matrix	Attention-only r32 (seed 13)	0.5739	87.55	84.27	**76.72**	54.60	78.28	92.88
OCR ablation	OCR ablation: on	0.5672	87.55	84.25	75.77	54.85	77.44	92.70
Core matrix	All-linear r16 (seed 07)	0.5803	87.45	84.20	76.23	54.71	77.87	93.25
Core matrix	Attention-only r32 (seed 07)	0.5798	87.30	83.65	76.17	54.72	77.70	93.60

Table 5: Post-train internal-dev results for the full LoRA study. Core-matrix rows compare target strategy, rank, and seed. Follow-up rows inherit the winner-conditioned family and probe OCR conditioning or data scaling. Bold entries mark the best metric value in the table, with lower being better for `eval_loss`.

Branch	Setting	Acc.	Cons.	F1	BLEU	METEOR	ROUGE-L	Judge
Prompt study	question routed	**75.82**	**71.41**	**67.95**	13.76	46.39	**71.45**	**87.25**
Stress test	generation cap 12	74.16	69.53	66.80	**15.53**	**47.25**	70.27	85.89
Prompt study	ocr copy first	74.00	69.55	67.20	14.55	47.11	70.95	87.01
Prompt study	baseline finalist	73.08	68.56	66.37	14.70	46.94	69.92	85.41
Stress test	ocr cap 16	73.02	68.53	66.33	14.66	46.89	69.94	85.71
Stress test	ocr cap 64	72.84	68.27	66.27	14.72	46.87	69.84	84.04
Prompt study	answer format constrained	71.66	67.17	64.72	13.94	45.11	68.46	83.97
Stress test	generation cap 4	64.76	60.97	61.76	7.86	43.24	65.87	80.21

Table 6: Validation robustness results from the appendix branch. These runs are intentionally secondary to the main funnel and are used to probe prompt-routing and decoding sensitivity without changing the core winner-selection logic. Bold entries mark the best value in each metric column.

alternative prompt formulations that enrich the interpretation, but it does not retroactively rewrite the finalist protocol. Among the stress tests, lowering the generation cap to 4 is clearly the worst setting under both deterministic and judge-based scoring, while more moderate caps remain much closer to the finalist baseline. The appendix therefore supports a nuanced conclusion: Qwen is reasonably robust to modest decoding perturbations, but overly aggressive truncation is materially harmful.

5 Analysis and Failure Modes

5.1 Where tuning helps most

The tuned model's gains are not evenly distributed across the validation set. Figure 3 already shows the broad pattern: adaptation helps most when the answer is recoverable from OCR tokens, especially after normalization. Compared with the strongest zero-shot finalist, the tuned model improves validation accuracy by 4.98 points for direct OCR matches and by 7.46 points for answers that become recoverable only after normalization. The gain is still positive when the answer is absent from OCR entirely (+5.57 points), but it is smaller than the gain on normalized OCR-grounded cases. This is a useful result for interpretation. Fine-tuning is not merely making the model more generally capable; it appears to improve how the model uses text that is already present but noisy, fragmented, or only approximately aligned with the canonical answer string.

The OCR-count buckets tell the same story at a different level of granularity. Validation accuracy on OCR-free cases is unchanged at 66.09%, whereas OCR-rich cases improve substantially: +4.52 points for 1–5 OCR tokens, +5.85 for 6–15, +8.46 for 16–30, and +9.09 for 31+ tokens. In other words, the strongest adaptation gains occur exactly where TextVQA is most text dependent. This makes the final model more convincing as a TextVQA system rather than merely as a stronger short-answer VQA model.

The answer-length analysis also sharpens the interpretation. The tuned model improves one-token answers by 5.85 points, 2–3-token answers by 6.31 points, and 4+-token answers by 7.09 points. Even

Question prefix	Count	Zero-shot	Tuned	Gain
what	3911	77.76	**83.38**	5.63
who	236	83.05	**85.59**	2.54
how	184	75.00	**83.70**	8.70
other	176	76.14	**88.64**	12.50
is	162	92.59	**99.38**	6.79
which	100	83.00	**92.00**	9.00
where	89	79.78	**93.26**	**13.48**

Table 7: Official validation accuracy by question prefix for prefix groups with at least 50 examples. Bold entries mark the better system within each prefix group and the largest observed gain.

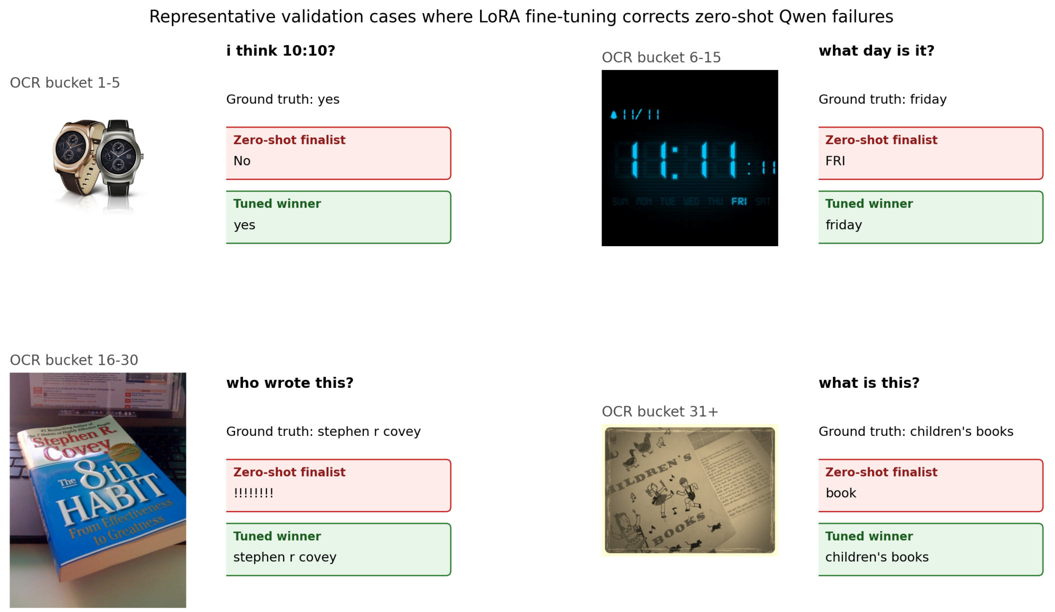

Figure 4: Representative validation cases where LoRA fine-tuning corrects failures made by the zero-shot Qwen finalist. The selected examples span OCR buckets from sparse text to dense text and illustrate three recurring recovery patterns: stricter answer-length control (top left), more faithful OCR grounding (top right), and better normalization or denoising of noisy book-cover text (bottom row).

after tuning, however, long answers remain the hardest category: 76.69% validation accuracy versus 85.63% for one-token answers. The gain is therefore real but not saturating. Parameter-efficient tuning helps on harder answer forms, yet does not eliminate the structural difficulty of longer free-form answers.

Question-prefix breakdowns point in the same direction while satisfying a more conventional per-category analysis. Table 7 shows that the tuned model improves every validation prefix group with at least 50 examples. The largest gains appear on `where`, `which`, `how`, and the heterogeneous `other` bucket, whereas `who` improves more modestly from an already strong zero-shot baseline. This pattern suggests that LoRA is not only correcting a narrow answer-format artifact; it improves several common question forms, with especially large gains where the answer often depends on locating or normalizing visible text.

5.2 Qualitative evidence

Figure 4 makes the qualitative pattern concrete. In the sparse-OCR watch example, the zero-shot finalist over-predicts the visible time string rather than answering the binary question, while the tuned model emits the short categorical answer required by the task. In the digital-clock example, zero-shot Qwen partially reads the day token but returns an abbreviated form that fails exact matching, whereas the tuned model normalizes it to the expected answer. The book-cover examples are even more revealing: the tuned model recovers author and title strings from cluttered or degraded text where the zero-shot model either hallucinates punctuation or collapses the answer to an underspecified noun. These cases are consistent with the bucket analysis above. The fine-tuned system is not simply more verbose or more aggressive; it is more disciplined in turning OCR evidence into task-compatible short answers.

5.3 Failure modes and limits

The study also exposes limits that remain after tuning. First, OCR-free questions remain a relative weakness. The tuned model does not improve the zero-shot finalist on the zero-OCR bucket, which suggests that the observed gains are genuinely tied to text-grounded reasoning rather than to a uniform shift in visual question answering quality. Second, despite strong overall improvements, the tuned system is still materially worse on longer answers and on some semantically ambiguous prompts. Third, the OCR ablation follow-up indicates that explicit OCR injection is not the sole driver of post-train gains; once the LoRA adapter is learned, the model appears to internalize part of the text-grounding benefit, but this also makes it harder to isolate exactly which adaptation parameters are compensating for OCR-format differences.

There is also an important methodological limit in the internal-dev split. As Table 1 reports, internal-dev is question-disjoint but not fully image-disjoint from the LoRA training pool. A post-hoc sensitivity audit partly bounds the issue: for the selected `all-linear-r16-seed13` adapter, internal-dev accuracy is 88.45% overall, 88.87% on image-overlap questions, and 87.39% on the 571 image-disjoint questions. This confirms that the internal-dev score is slightly higher on overlapping images, so we do not treat internal-dev as final held-out evidence. It also shows that the selected adapter remains strong on the image-disjoint subset, and the final claim is ultimately based on the official validation result rather than on internal-dev alone.

Although the final paper includes API-backed LLM-as-a-Judge scores across the reported experiment tables, we do not use that score as a global replacement for benchmark-native exact match. The judge is most useful as a secondary semantic lens: it tells us whether leaderboard differences survive a more tolerant reading of short answers and whether exact-match gaps are mostly semantic or mostly formatting driven. That is worthwhile additional evidence, but it does not eliminate the need for deterministic, reproducible benchmark metrics across the rest of the study.

Finally, the adaptation story is intentionally narrow. We tune only the selected Qwen backbone and only with LoRA. That decision is scientifically coherent with the funnel design and with the available compute budget, but it leaves open at least two follow-up questions: whether the same winner-conditioned protocol would select a different adaptation family on a larger backbone, and whether a richer parameter-efficient method would preserve the same OCR-grounded gains without increasing instability or cost.

6 Conclusion

This paper presented a staged TextVQA study that was designed to be broader than a single tuned-model report while remaining methodologically disciplined under a realistic compute budget. The zero-shot funnel established Qwen2.5-VL-3B as the strongest open-source backbone in the benchmark pool, the finalist reruns showed that this conclusion survived transfer to the official validation split, and the winner-conditioned LoRA study converted that backbone advantage into a tuned validation score of 84.76%. The resulting paired gain over the strongest zero-shot finalist is 6.04 points on the same 5,000 official validation questions, with a bootstrap 95% confidence interval of $[5.00, 7.04]$ points. The main empirical conclusion is therefore anchored on official validation evidence, while the internal-dev split is used for screening and diagnostics.

The empirical findings are also more specific than a single leaderboard improvement. Prompting matters, but mostly within a backbone rather than across backbones. Fine-tuning helps, but it helps most on OCR-grounded questions rather than on OCR-free ones. Training-stage loss is a useful signal for allocating follow-up budget, but not a substitute for answer-accuracy-based promotion or official validation evaluation. OCR toggling matters less after adaptation than it does in zero-shot prompting, while additional training data appears to help modestly in the diagnostic follow-up runs. Together, these observations make the final tuned model more interpretable than a generic "best checkpoint" claim.

From a project-design perspective, the work also illustrates a practical template for academic multimodal studies. The key ingredients are not exotic: a stratified internal-dev split with an explicit split audit, a clear promotion rule, a hard boundary between local evaluation and remote training, a winner-conditioned follow-up phase, and artifact-level reproducibility. Yet in combination, these decisions make the final evidence substantially easier to defend. Future extensions should add a semantically richer learned judge with human calibration, expand adaptation beyond a single backbone, and test whether the same funnel protocol transfers to adjacent OCR-heavy benchmarks. Even without those extensions, the current study supports a clear conclusion: carefully adapted open-source VLMs can become materially stronger TextVQA systems when the experimental protocol separates screening evidence from final held-out validation evidence.

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